9 A. Anatropous Ovule

Diagnostic Features

- The body of the ovule becomes completely inverted so that micropyle lies close to the funicle.
- Micropyle and chalaza lie on the same straight line. Example: Asteraceae.

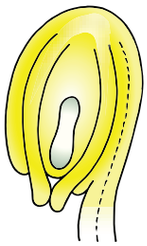


Figure 8a: Anatropous ovule

9 B. Orthotropous Ovule

Diagnostic Features

- In this type of ovule, the micropyle is at the distal end.
- The ovule is erect or straight so that the funicle, chalaza and micropyle lie on the same vertical line. Example: Piperaceae and Polygonaceae.

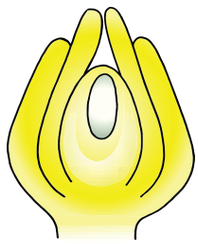


Figure 8b: Orthotropous ovule

9 C. Campylotropous Ovule

Diagnostic Features

- In this type, the body of the ovule at the micropylar end is curved and more or less bean shaped.
- The embryosac is slightly curved.
- The funicle, micropyle and chalaza are adjacent to one another with the micropyle oriented towards the placenta. Example: Leguminosae.

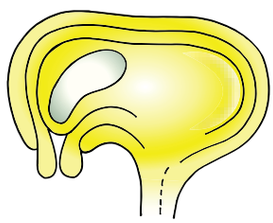


Figure 8c: Campylotropous ovule

Exercise 10: E.coli cloning vector (pBR 322)

Aim: To study and identify the features of cloning vector – pBR 322

Principle: Vectors are used as carriers to deliver the desired foreign DNA into a host cell.

Requirements: Models/ Photographs / Pictures of E.coli Cloning vector pBR 322.

Diagnostic Features

- pBR 322 plasmid is a reconstructed plasmid containing 4361 base pairs and most widely used as cloning vector.
- In pBR, p denotes plasmid and B and R respectively the notes of scientists Boliver and Rodriguez who developed the plasmid. The number 322 is the number of plasmids developed from their laboratory.
- It contains two different antibiotic resistance genes and recognition site for several restriction enzymes (Hind III, Eco R I, Bam H I, Sal I, Pvu II, Pst I, Cla I), Ori and antibiotic resistance genes (amp^R and tet^R). Rop codes for the proteins involved in the replication of the plasmid.

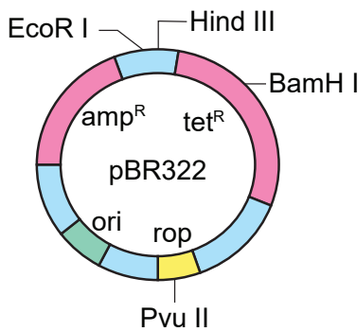


Figure 10: E-coli cloning vector (pBR 322)

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